

Setting up the EEG software

1. Download [environment.yml file](#)
2. Open your Miniconda command prompt
3. Within the command prompt, browse to your downloads folder using the cd command:

```
cd /Users/Yourname/Downloads
```

5. Then copy and paste the command:

```
conda env create --name mini_eeg --file=eeg_environment.yml
```

6. Copy and paste the command:

```
conda activate mini_eeg
```

7. Now run the EEG recorder (which should also be in your Downloads folder):

```
python eegrecorder.py
```

Running the program

1. In the EEG recorder, enter a number of "Samples" in the white box next to Record, each Sample takes one second, so enter 50 to get about a minute of live electrical recording
2. Hit "Plot" to get live recording
3. Hit "Record" to get live recording that will then be saved into your Downloads folder as a CSV (this contains the data that was being plotted for your own use/analysis). The file is called eeg_recording.csv

Saving the data. Each time you hit record, it will overwrite the eeg_recording.csv file. So if you wanted to record one CSV with eyes open and one CSV with eyes closed, follow these steps:

1. Enter the number of samples (e.g. 50)
2. Push "Record"
3. Keep your eyes open
4. Once recording is done, go to Downloads folder, rename the file eeg_recording.csv to eeg_recording_open.csv
5. Return to the EEG recorder and once again push "Record"

6. Keep your eyes closed
7. Once recording is done, go to Downloads folder, rename the file eeg_recording.csv to eeg_recording_closed.csv
8. Now you have the data of your brain waves in the two situations!

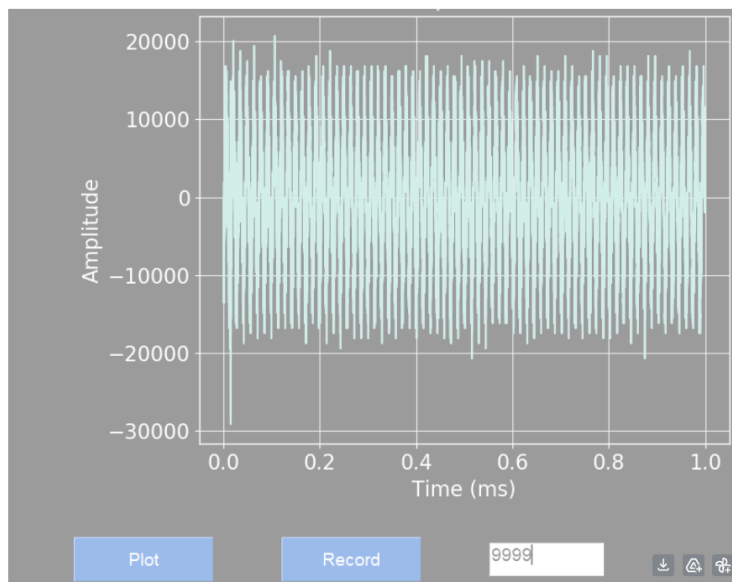
Troubleshooting

- If the program does not open up and it says something like:

```
Traceback (most recent call last):
  File "eegrecorder.py", line 247, in <module>
    app=Application(master=root)
  File "eegrecorder.py", line 154, in __init__
    self.createWidgets()
  File "eegrecorder.py", line 193, in createWidgets
    self.plot(canvas,ax1,ax2,inputtxt)
  File "eegrecorder.py", line 201, in plot
    output = plotandrecord()
  File "eegrecorder.py", line 110, in plotandrecord
    times, eegdata = livestream()
  File "eegrecorder.py", line 97, in livestream
    data_all = trim(data_all) # we trim before normalize as threshold applies to un-normalized wave (as well as is_sil
ent() function)
  File "eegrecorder.py", line 70, in trim
    to_return = copy.deepcopy(data_all[_from:(_to + 1)])
TypeError: slice indices must be integers or None or have an __index__ method
```

It means the computer is not finding an input. Check that all the cables are connected and the EEG is powered on with a charged battery

- Occasionally, if you hit the Plot button more than once, the program will freeze and you can tell because the screen becomes a pale shade.



In this case, you should hit "X" or "Ctrl-Alt-Delete" to Force quit the program